

32-bit microcontrollers

AES module

Applicable to:

Series	Product Model	Series	Product Model
HC32L13	HC32L130E8PA	HC32F03	HC32F030E8PA
	HC32L130F8UA		HC32F030F8UA
	HC32L130J8TA		HC32F030F8TA
	HC32L136J8TA		HC32F030H8TA
	HC32L136K8TA		HC32F030J8TA
			HC32F030K8TA
HC32L07	HC32L072PATA	HC32F07	HC32F072PATA
	HC32L072KATA		HC32F072KATA
	HC32L072JATA		HC32F072JATA
	HC32L073PATA		
	HC32L073KATA		
	HC32L073JATA		
HC32L17	HC32L176PATA	HC32F17	HC32F176PATA
	HC32L176MATA		HC32F176MATA
	HC32L176KATA		HC32F176KATA
	HC32L176JATA		HC32F176JATA
	HC32L170JATA		HC32F170JATA
	HC32L170FAUA		HC32F170FAUA
HC32L19	HC32L196PCTA	HC32F19	HC32F196PCTA
	HC32L196MCTA		HC32F196MCTA
	HC32L196KCTA		HC32F196KCTA
	HC32L196JCTA		HC32F196JCTA
	HC32L190JCTA		HC32F190JCTA
	HC32L190FCUA		HC32F190FCUA

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1. Abstract

This application note primarily introduces the use of the AES module in Huada Semiconductor MCUs*.

This application note includes:

- Register Operations
- Encryption
- Decryption
- Precautions
- Exception Mechanism
- Performance

Note:

- This application note is supplementary material for Huada Semiconductor MCUs* and cannot replace the user manual. For specific functions and register operations, please refer to the user manual.

2. Function Introduction

This article explains the application of the AES module in Huada Semiconductor's MCU*.

- It executes the encryption and decryption processes of the AES algorithm standard, and its execution results fully comply with the description of the algorithm principle in FIPS PUB 197;
- It only supports 128-bit keys.

****Supported models are listed on the cover.***

3. AES Module

3.1 Register Operations

The data register consists of four 32-bit registers, providing 128 bits of data.

It is used to store plaintext to be encrypted or ciphertext to be decrypted before the module performs operations, and to store the encrypted ciphertext or decrypted plaintext after the operations are completed.

Encryption Operation		Decryption Operation	
Before Operation	After Operation	Before Operation	After Operation
128-bit Plaintext	128-bit Ciphertext	128-bit Ciphertext	128-bit Plaintext

Four 32-bit registers are connected together to form a 128-bit data set. Read and write operations require separate operations on each of the four registers. The operation sequence for the data registers is as follows:

Example data: 0x00112233445566778899AABBCCDDEEFF

Data0	0x33221100
Data1	0x77665544
Data2	0xBBAA9988
Data3	0xFFEEDDCC

The key register consists of four 32-bit registers, storing the initial input key.

Write operations require manipulating each of the four 32-bit registers separately. The corresponding operation sequence is as follows:

Example data: 0x000102030405060708090A0B0C0D0E0F

Key0	0x03020100
Key1	0x07060504
Key2	0x0B0A0908
Key3	0x0F0E0D0C

- Writes to the data and key registers can only be performed when this module is not in operation (i.e., CR.Start = 0). Otherwise, the hardware will automatically ignore write operations to these registers.
- Reads from the data and key registers can only be performed when this module is not in operation (i.e., CR.Start = 0). Otherwise, reading from these registers will return all zeros.

3.2 Encryption

- Write the 128-bit data to be encrypted into the data register (DATA).
- Write the encryption key into the key register (KEY).
- Set CR.Mode to 0.
- Write 1 to CR.Start in the control register to start the module's operation.
- Wait for the value of CR.Start to return to 0; the module's operation ends.
- CR.Mode and CR.Start can be configured simultaneously.
- Read the data register (DATA) to obtain the 128-bit ciphertext.

3.3 Decryption

- Write the 128-bit data to be decrypted into the data register (DATA).
- Write the decryption key into the key register (KEY).
- Set CR.Mode to 1.
- Write 1 to CR.Start in the control register to start the module's operation.
- CR.Mode and CR.Start can be configured simultaneously.
- Wait for the value of CR.Start to return to 0; the module's operation ends.

3.4 Precautions

- After power-on, the clock hclk must be stable and valid before being released from reset, and remain stable throughout subsequent operation.
- During AES encryption and decryption, the data register changes. If the operand of the next operation is the result of the current operation, then there is no need to rewrite the data.
- The key only supports 128 bits, and the key is written to offset addresses 0x20-0x2C.
- The method for determining the end of the module operation: continuously read CR.Start; if its value becomes 0, it indicates that the operation has ended.

3.5 Exception Handling

- Only 32-bit access is supported; access to other bit widths will return a hardware fault.
- Accessing data and keys at addresses with offsets greater than 0x0F will also return a hardware fault.

3.6 Performance

The time required for this module to start an operation (CR.Start is written to 1) and for the operation to end (CR.Start is restored to 0).

encryption	216 cycles
Decrypt	286 cycles

4. Reference Examples and Drivers

Through the above introduction, and in conjunction with this series of user manuals, we have gained a deeper understanding of the functions and operation methods of the AES module in Huada Semiconductor's MCU*.

Huada Semiconductor (HDSC) also provides application examples and driver libraries for this module. Users can open the example projects to further familiarize themselves with the application of the module and driver library intuitively. In actual development, users can also directly refer to the examples and use the driver library to quickly implement operations on this module.

5. Summary

The preceding sections briefly introduced the AES module of Huada Semiconductor's MCU*, and explained its usage in detail. For a more in-depth understanding of the module's usage and operation during actual application development, users should refer to the corresponding user manual. The examples and driver libraries mentioned in this article can be used for further experimentation and learning, or directly applied in actual development.

**Supported models are listed on the cover.*

6. Other Information

Technical support information: www.hdsc.com.cn

7. Version Information & Contact Us

Date	Version	Change History
2018/6/20	Rev1.0	Initial release.
2018/9/3	Rev1.1	Updated supported product models.
2019/6/24	Rev1.2	Updated supported product models.



If you have any comments or suggestions during the purchase and use process, please feel free to contact us.

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